



CHEMISTRY HSSC-I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر غلطی کے حوالے کریں۔ کاٹ کر دہرا لکھنے کی اجازت نہیں ہے۔ سب سے پہلے اس سے متبادل منسوخ ہے۔

Version No.				
3	2	0	5	1

ROLL NUMBER						

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The electronic configuration of an ion with mass number 63 and charge +2 is $[Ne] 3s^2, 3p^6, 3d^9$. The number of electrons, protons and neutrons present in the atom are:	e=27 p=27 n=36	e=29 p=27 n=36	e=29 p=29 n=34	e=34 p=29 n=63	☐	☐	☐	☐
2. A molecule with formula H_2CO has three sigma bond pairs and one pi bond pair in valence shell of central atom (carbon). Its structure is:	Triangular pyramid	Trigonal planar	Bent	Tetrahedral	☐	☐	☐	☐
3. $0.5 dm^3$ of CO_2 gas is enclosed in a container at STP. The mass of gas is:	0.49g	0.98g	22.4g	48.6g	☐	☐	☐	☐
4. Which molecule possesses stronger London dispersion forces?	n-Butane	n-Pentane	Iso butane	Iso pentane	☐	☐	☐	☐
5. Which of the following is the correct relationship for Gibb's free energy?	$G = H + TS$	$G = H \times TS$	$G = H - TS$	$G = \frac{H}{TS}$	☐	☐	☐	☐
6. Correct rate equation for given reaction is: $H_2 + Cl_2 \xrightarrow{\text{sunlight}} 2HCl$	Rate = $k[H_2][Cl_2]$	Rate = $k[H_2]$	Rate = $k[Cl_2]$	Rate = k	☐	☐	☐	☐
7. Which of the following reaction at equilibrium will NOT be affected by change in pressure?	$N_2O_4 \rightleftharpoons 2NO_2$	$CO + 2H_2 \rightleftharpoons CH_3OH$	$N_2 + O_2 \rightleftharpoons 2NO$	$N_2 + 3H_2 \rightleftharpoons 2NH_3$	☐	☐	☐	☐
8. If solubility of a salt MX is $2 \times 10^{-3} \text{ mol/dm}^3$, what will be the value of K_{sp} if salt dissociates into one cation (M^+) and one anion (X^-)?	1×10^{-4}	2×10^{-4}	4×10^{-4}	8×10^{-4}	☐	☐	☐	☐
9. Which of the following decreases across the period?	Atomic radius	Ionization potential	Electron affinity	Electronegativity	☐	☐	☐	☐
10. An element M reacts with O_2 , H_2O and Cl_2 to form the products M_2O , MOH and MCl respectively. The element is most likely:	Li	Al	Mg	Ca	☐	☐	☐	☐
11. Which compound of sulfur is used as fertilizer?	$(NH_4)_2SO_4$	$CuSO_4$	Na_2SO_4	$BaSO_4$	☐	☐	☐	☐
12. Identify x and y, in the reaction: $CH_2 = CH_2 \xrightarrow{x} C_2H_5OH \xrightarrow{y} CH_3 - CH_2 - OH$	x = H_2O , y = HCl	x = H_2SO_4 , y = $LiAlH_4$	x = H_2SO_4/H_2O , y = Al_2O_3	x = $NaBH_4$, y = H_2SO_4	☐	☐	☐	☐
13. An alkyl halide is reacted with $AgNO_3$ in ethanol, light yellow ppt insoluble in NH_4OH are formed. Identify the alkyl halide:	$CH_3 - CH_2 - F$	$CH_3 - CH_2 - Cl$	$CH_3 - CH_2 - Br$	$CH_3 - CH_2 - I$	☐	☐	☐	☐
14. What is produced when ethene is reacted with cold, dilute solution of $KMnO_4$?	$CH_3 - CH_2 - OH$	$CH_3 - CHO$	$CH_3 - COOH$	$HO - CH_2 - CH_2 - OH$	☐	☐	☐	☐
15. Identify the product Q in given reaction: $CH_3 - CH_2 - CN + 2H_2O \xrightarrow{HCl} Q + NH_3$	$CH_3 - CH_2 - OH$	$CH_3 - CH_2 - COOH$	$CH_3 - CH_2 - NH_2$	$CH_3 - COOH$	☐	☐	☐	☐
16. Butanone can be prepared by oxidation of:	1-Butanol	2-Butanol	Butanal	Butanoic acid	☐	☐	☐	☐
17. The reduction of $C_6H_5 - NO_2$ by Sn/HCl results in the formation of:	C_6H_6	$C_6H_5 - NH_2$	$C_6H_5 - OH$	$C_6H_5 - Cl$	☐	☐	☐	☐

—1HA-I 25005 (D)—

SUPPLEMENTARY TABLE

Page 1 of 1

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40



CHEMISTRY HSSC-I

28

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly.

(14 x 3 = 42)

(i)	Write down the electronic configuration of an element with atomic number 24 and also of its ion with charge +3. Describe the reason for any discrepancy in electronic configuration of element.	1x3	OR	Briefly explain the terms with examples: a) Enthalpy of atomization, b) Lattice energy, c) Electron affinity.	03
(ii)	Draw molecular orbital diagram for N_2 and calculate its bond order.	03	OR	What is dynamic equilibrium? How does it exist between two physical states of a substance? Describe by giving an example.	1+2
(iii)	What is molar volume? Calculate the volume occupied by 2000g of Cl_2 gas at STP.	1+2	OR	How are the elements arranged in different blocks of periodic table?	03
(iv)	Why does carbon form a large number of organic compounds? Give three reasons.	03	OR	Complete the following reactions: i $CH_2 = CH_2 + H_2O + KMnO_4 \longrightarrow$ ii $CH_3 - CH_2 - COOCH_3 + NaOH \longrightarrow$ iii $C_6H_5CHO + 2[H] \xrightarrow{NaBH_4}$	03
(v)	What is Azimuthal quantum number? Describe its significance.	1+2	OR	Explain why: i is the falling drop of water spherical in shape? ii has water high surface tension than Ether?	1.5 + 1.5
(vi)	Justify the difference in the boiling points of $CH_4(-162^\circ C)$, $HCl(-85^\circ C)$ and $H_2O(100^\circ C)$.	03	OR	What is the Photo chemical smog? Briefly describe its formation.	1+2
(vii)	What is entropy? Describe the increase or decrease in entropy of system with sign in the following reactions: a) $N_2O_{4(g)} \longrightarrow 2NO_{2(g)}$ b) $NH_{3(g)} + HCl_{(g)} \longrightarrow NH_4Cl_{(s)}$	1x3	OR	Ethanol can be converted into propanoic acid in three stage process. $C_2H_5OH \xrightarrow{I} A \xrightarrow{II} B \xrightarrow{III} C_2H_5 - COOH$ Write down the balanced chemical equation to show the conversion in each step.	03
(viii)	Write down three methods by which un-useable water can be converted into clean water.	03	OR	Write down any three harmful effects of acid rain.	03
(ix)	Write down three methods for the preparation of Ethyl chloride from an alcohol.	03	OR	How can primary, secondary and tertiary alcohols be distinguished by applying Lucas test?	03
(x)	Briefly explain the factors which affect the relative stability of primary, secondary and tertiary carbocation.	03	OR	Write down the reactions of Ethyl Amine ($C_2H_5 - NH_2$) with: a) $CH_3CH_2 - Cl$ b) HNO_2 / HCl	1.5 + 1.5
(xi)	Differentiate between metallic and non-metallic behavior of elements.	03	OR	For the reaction $2NO_2 + F_2 \longrightarrow 2NO_2F$ $Rate = [NO_2][F_2]$ propose the mechanism of reaction.	03
(xii)	Describe how $(n+l)$ rule and Hund's rule are applied in writing the electronic and orbital configuration of $Mn (Z=25)$.	03	OR	How does a catalyst affect the rate of a chemical reaction? Explain by drawing a reaction pathway diagram.	03
(xiii)	Calculate concentration of H^+ ion in 1.0 M solution of Formic acid ($HCOOH$) which ionizes as: $HCOOH \rightleftharpoons H^+ + HCOO^-$, $K_a = 1.8 \times 10^{-4}$	03	OR	How can Iodoform test be applied for identification of aldehydes and ketones? Describe by giving chemical equations.	1.5 + 1.5
(xiv)	Draw position isomers and functional group isomers of $C_3H_7 - OH$	1.5 + 1.5	OR	Why is the ionization of HF decreased when aqueous solution of its salt is added to it? Briefly explain by applying Le-Chatlier's principle.	03

SECTION – C (Marks 26)

Attempt the following questions.

Q.3	What is hybridization? Write down three differences between sp^3 and sp^2 hybridizations. Explain bonding in BF_3 on the basis of hybridization.	1+3 +3	OR	Why are alkanes chemically less reactive? Describe the mechanism of free radical chlorination of Ethane to form Ethyl Chloride.	3+4
Q.4	Explain one physical and one chemical method to recognize a system at equilibrium.	3+3	OR	Write down a chemical reaction between CH_3COOH and CH_3OH with necessary conditions. Also write the mechanism of this reaction, explaining each step.	2+4
Q.5	Explain the relative reactivity of halogens: (i) as oxidizing agents (ii) in their reaction with H_2 , by giving suitable examples.	3+3	OR	Describe the behaviour of 1-propanol and 2-propanol with respect to their: (i) reaction with $I_2/NaOH$ (ii) reaction with $HCl/ZnCl_2$ (iii) reaction with $K_2Cr_2O_7/H_2SO_4$	2x3
Q.6	300 g Zn is reacted with 300 g HCl . $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$ Calculate the mass of H_2 produced. In an experiment, if 6g H_2 is obtained, calculate percentage yield. (Atomic masses $Zn=65.38$ g/mol $H=1$ g/mol $Cl=35.5$ g/mol)	07	OR	State Hess's law with an example. Apply Hess's law to calculate enthalpy of formation of CH_3COOH . The standard enthalpies of combustion of carbon, hydrogen and acetic acid are -393.5 kJ/mol, -285.8 kJ/mol and -875 kJ/mol respectively.	2+5

SUPPLEMENTARY TABLE

— 1HA-I 25005 (D) —

Atomic No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Symbol	H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar	K	Ca
Mass No	1	4	7	9	11	12	14	16	19	20	23	24	27	28	31	32	35.5	40	39	40